



**Linux Cluster Monitoring Agent
Project #: 01
Business Requirements Document**

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Executive Summary

Our organization is initiating the Linux Cluster Monitoring project to implement a centralized system for collecting and tracking hardware resource usage across all servers in the data center. The primary goal of this project is to provide the Linux Cluster Administration (LCA) team with consistent, accurate visibility into cluster performance, enabling informed decisions on server capacity, resource allocation, and long-term infrastructure planning.

The monitoring solution will capture key host and usage metrics from each server and consolidate them into a central repository. This will support the generation of daily operational reports and lay the groundwork for future analytical capabilities. By standardizing data collection and reporting, the project addresses core business needs such as enhanced operational insight, improved capacity planning, and consistent performance monitoring.

Multiple stakeholder groups will participate, including the Project Manager, LCA team, Implementation (technical) team, and Support team, each contributing to the design, deployment, and adoption of the solution.

The project is scheduled for January 2026, with completion targeted for January 31, 2026, in alignment with organizational training and operational planning cycles. This BRD defines the business objectives, requirements, scope, stakeholders, timeline, and cost-benefit considerations associated with the initiative.

Project Objectives

The primary objectives of the Linux Cluster Monitoring project are as follows:

1. Deploy a monitoring agent across all Linux cluster servers to capture and persist hardware metrics.
2. Enable the LCA team to access daily reports summarizing cluster utilization.
3. Ensure that data collection is automated, consistent, and scalable across all cluster nodes.
4. Provide knowledge transfer and documentation to ensure smooth adoption and ongoing use of the system.

The project aligns with the organization's goal of optimizing infrastructure usage and reducing manual effort associated with hardware monitoring.

Needs Statement

Currently, the Linux Cluster Administration (LCA) team collects hardware and usage metrics manually by accessing each server individually to check CPU, memory, and disk utilization. This approach is time-consuming, inconsistent, and increasingly difficult to scale as the cluster expands. Because data is gathered irregularly and stored in varying formats, the team lacks a reliable, consolidated view of server health, which limits proactive capacity planning and often relies on outdated information.

A centralized monitoring system is required to automate data collection, minimize errors, and provide accurate, up-to-date insights into cluster performance. By capturing and consolidating hardware usage metrics automatically, the system will enable the LCA team to make faster, more informed decisions regarding resource allocation, provisioning, and long-term infrastructure planning.

Project Scope

The Linux Cluster Monitoring initiative is focused on implementing a streamlined solution to automatically collect, centralize, and report hardware usage metrics from Linux servers. This project will cover the development, deployment, and configuration of monitoring components within the organization's internal Linux cluster. All activities—including testing, reporting, and documentation—are limited to the internal environment.

In Scope:

- Deploying a monitoring agent on each Linux server to gather both static and dynamic hardware metrics.
- Creating a PostgreSQL database (running in Docker) to store and manage collected data.
- Scheduling automated data collection through cron jobs at defined intervals.
- Producing daily hardware usage reports for the LCA team.
- Performing validation and testing in development and production environments.
- Preparing documentation and delivering knowledge transfer to support system operations and maintenance.

Out of Scope:

- Monitoring non-Linux servers, cloud-based infrastructure, or hybrid environments.
- Providing advanced analytics, trend forecasting, dashboards, or visualization tools.
- Integrating with existing enterprise monitoring platforms or third-party tools.

Requirements

The Linux Cluster Monitoring system must consistently gather, store, and report hardware resource metrics from all nodes within the cluster. Requirements are categorized into business-level needs and the technical specifications necessary to meet them.

Business Requirements:

- Provide reliable and timely visibility into hardware usage to support capacity planning and efficient resource allocation.
- Automatically generate daily reports for the LCA team, eliminating the need for manual data collection.
- Record static host information once per server for reference and historical purposes.
- Collect hardware usage data at regular, defined intervals to ensure consistency and integrity.

Technical Requirements:

- Deploy a Bash-based monitoring agent capable of retrieving CPU, memory, disk, and system metadata from each server.
- Maintain a centralized PostgreSQL database hosted in Docker for persistent storage of collected metrics.
- Use cron jobs to automate routine data collection from all servers.
- Capture key metrics including CPU utilization, CPU kernel time, available memory, disk I/O, disk capacity, and host hardware details.

- Provide SQL queries and scripts to generate daily cluster usage reports.
- Include comprehensive documentation covering installation, configuration, and ongoing system maintenance.

Key Stakeholders

Project Activity	Project Sponsor	Project Manager	LCA Team	Implementation / Agile Team	Support Team
Define project objectives & scope	A	R	C	C	I
Gather business requirements	I	R	A	C	I
System architecture & design	I	C	C	A/R	I
Monitoring agent development	I	I	C	A/R	I
Database setup & configuration	I	C	I	A/R	C
Testing (DEV, QC/TEST, PRE-PROD)	I	C	C	A/R	I
Deployment to production servers	I	C	I	R	A
Documentation & knowledge transfer	I	R	C	A	C
Daily reporting implementation	I	C	A	R	I
System maintenance & support	I	I	C	C	A/R
Project approval & closure	A	R	I	I	I

Project Schedule & Milestone

Milestone	Description	Target Date
Requirements Gathering	Engage with all stakeholders to collect, review, and confirm business requirements.	Jan 16, 2026
MVP Completion	Develop the initial monitoring agent and design the database schema. Validate successful data collection using designated test hosts.	Jan 19, 2026
System Deployment	Deploy monitoring agent scripts and the PostgreSQL Docker instance across the internal cluster environment.	Jan 21, 2026
Testing	Execute comprehensive end-to-end testing across DEV, QC/Testing, pre-production and PROD environment to ensure system reliability and accuracy.	Jan 23, 2026
Knowledge Transfer	Deliver training sessions and provide complete documentation to the LCA team to support ongoing operations and maintenance.	Jan 26, 2026
MCR Review	Produce the first complete daily usage reports and conduct the final Master Code Review (MCR) of the full implementation.	Jan 27, 2026
Post-MCR Refinements	Incorporate final refinements, update documentation as needed, and complete project closure activities.	Jan 30, 2026

Cost–Benefit Analysis

Estimated Costs

- **Design and Deployment Effort:** The initiative requires dedicated time and resources to architect, develop, and implement the monitoring agent, data workflow, storage

infrastructure, and reporting mechanisms.

- **Quality Assurance and Verification:** Additional effort will be necessary to validate data integrity, test automated processes, and confirm stable performance across DEV, QC, and pre-production environments.
- **Documentation and Enablement:** Time must be allocated to create comprehensive user manuals, technical documentation, and to conduct knowledge-sharing sessions with the Linux Cluster Administration (LCA) team.
- **Ongoing Support and Optimization:** The solution will require periodic maintenance, minor script enhancements, and environment health checks to maintain long-term stability and performance.

Expected Benefits

Benefit Area	Current Situation	Expected Improvement	Measurable Outcome
Reduced Manual Work	Administrators manually collect metrics from each server.	Automated monitoring across all nodes.	85-90% reduction in manual administrative tasks.
Faster Operational Insights	Cluster performance analysis requires manual checks and delayed reporting.	Daily automated reporting of hardware metrics.	40-50% faster identification of resource utilization issues.
Improved Data Accuracy	Manual collection increases risk of inconsistencies and missing data.	Automated metric collection ensures standardized reporting.	90-95% improvement in data consistency.
Better Capacity Planning	Planning decisions rely on incomplete or outdated information.	Centralized historical data supports trend analysis.	25-35% improvement in capacity planning accuracy.
Operational Efficiency	Administrators spend time logging into multiple servers.	Centralized database and automated reporting.	30-40% reduction in monitoring effort.
Future Scalability	Limited ability to expand monitoring capabilities.	Architecture supports additional analytics and visualization tools.	Enables 100% cluster monitoring coverage and future enhancements.

Overall Return on Investment (ROI)

The anticipated advantages—especially the decrease in manual workload and the enhancement of resource planning—offer significant operational impact when compared to the relatively low development and maintenance expenses. Once deployed, the solution is expected to begin generating measurable efficiency improvements within a short timeframe.